DO NOW — RECALL

# Search engine indexing and PageRank

Name: _____ Class: _____ Date: _____

**1. What is a hyperlink?**

**2. What is an index in retrieval?**

**3. What kind of graph models links?**

**4. BST root20, left10, right30; insert 25. Where is it placed?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A reference to another page/resource.

### 2. Model answer

A structure associating terms or keys with documents/records.

### 3. Model answer

A directed graph.

### 4. Model answer

25>20 go right to30; 25<30 go left of30.

### Vocabulary — know these terms

crawler — Discovers/fetches documents.

inverted index — Maps terms to documents.

PageRank — Iterative link-based importance model.

damping — Probability factor in random-surfer model.

out-degree — Number of directed outgoing links.

NEW LEARNING

# Read, annotate and explain

### Understand

A crawler discovers pages by following links and sitemaps; an index records useful information about documents for searching.

### Apply

PageRank models a random web surfer and passes importance through directed links. A link from a high-ranked page can carry more weight than one from a low-ranked page.

### Evaluate

PageRank uses damping and repeated iterations; raw incoming link counts alone are not the full algorithm. Search engines combine many ranking signals.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

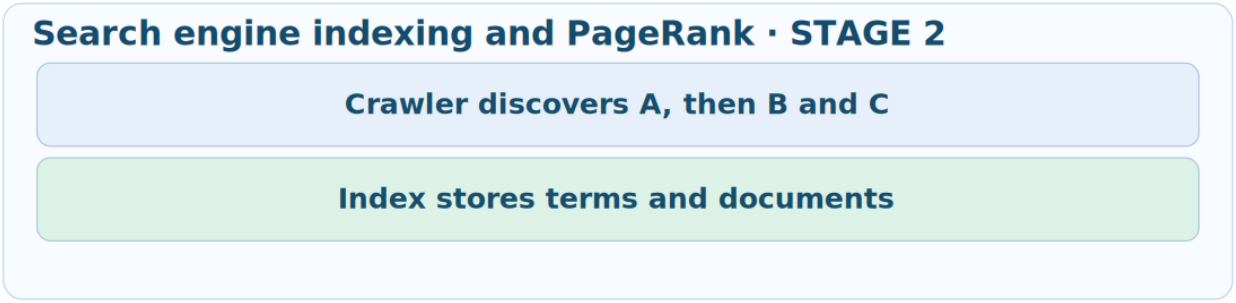
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



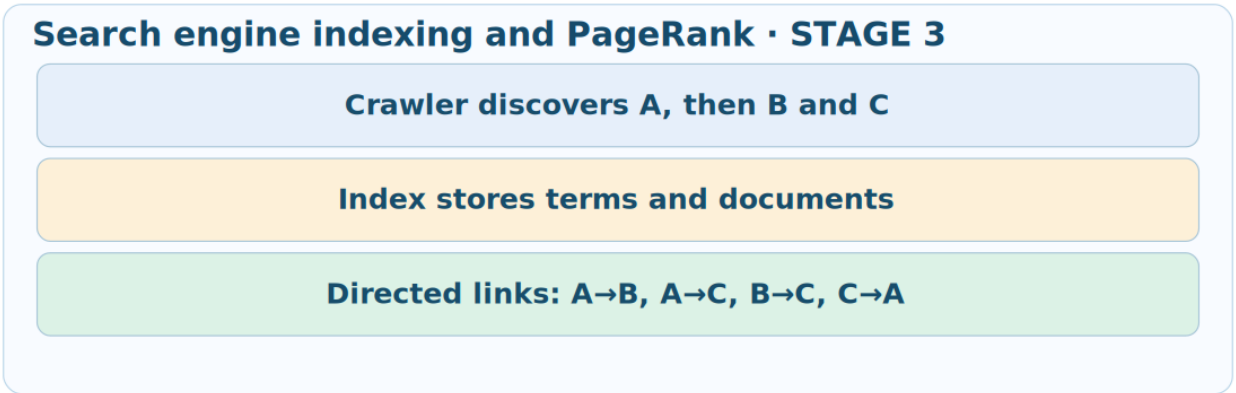

Search engine indexing and PageRank · STAGE 1; Crawler discovers A, then B and C

**New Learning**




Search engine indexing and PageRank · STAGE 2; Crawler discovers A, then B and C then Index stores terms and documents

**New Learning**

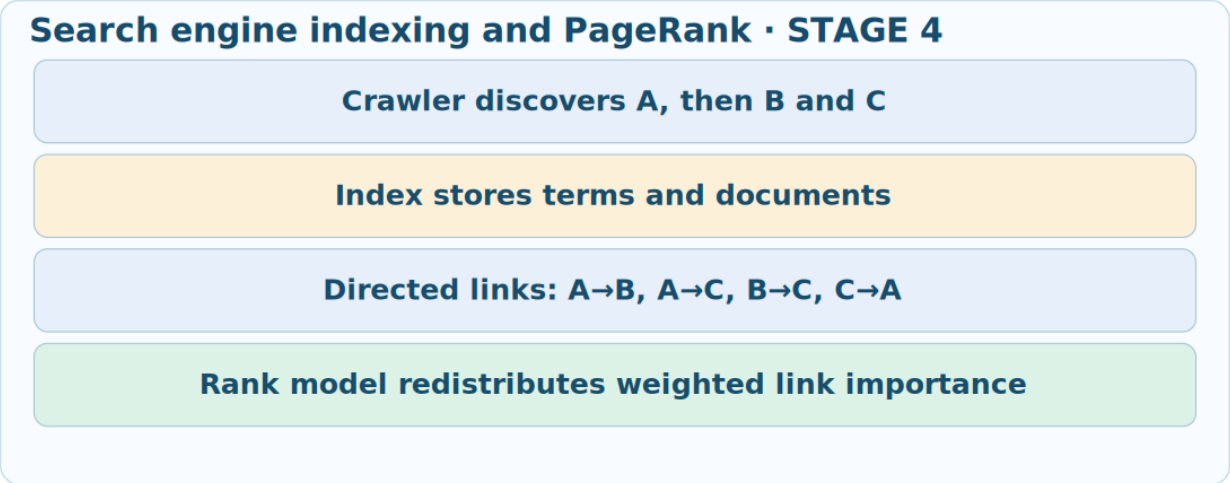



Search engine indexing and PageRank · STAGE 3; Crawler discovers A, then B and C then Index stores terms and documents then Directed links: A→B, A→C, B→C, C→A

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Search engine indexing and PageRank · STAGE 4; Crawler discovers A, then B and C then Index stores terms and documents then Directed links: A→B, A→C, B→C, C→A then Rank model redistributes weighted link importance

I DO — WORKED EXAMPLE

# Follow the worked method

### In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

### Predict first

Outgoing degree of A is two, so its rank is divided across two outgoing edges.

### Explain

Both B and C receive half of A's contribution before damping and other link contributions.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

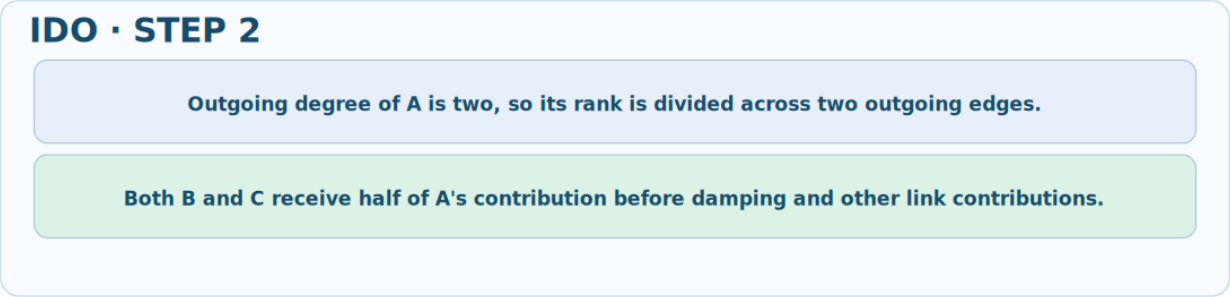
Outgoing degree of A is two, so its rank is divided across two outgoing edges.




IDO · STEP 1; Outgoing degree of A is two, so its rank is divided across two outgoing edges.

### In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

Outgoing degree of A is two, so its rank is divided across two outgoing edges.




IDO · STEP 2; Outgoing degree of A is two, so its rank is divided across two outgoing edges. then Both B and C receive half of A's contribution before damping and other link contributions.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A has three outgoing links to B,C,D and rank 0.30. What raw contribution reaches C?

The outgoing links split A's contribution evenly in the simplified model.




WEDO · STEP 1; The outgoing links split A's contribution evenly in the simplified model.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first The outgoing links split A's contribution evenly in the simplified model.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Distinguish crawling and indexing. [4]

### Question 2

In directed link graph A→B,C and B→C, identify incoming links of C and A's share given rank 0.4, ignoring damping. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why the number of inbound links alone is not an adequate PageRank calculation. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A page's PageRank is simply its number of incoming links.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A has three outgoing links to B,C,D and rank 0.30. What raw contribution reaches C?

The outgoing links split A's contribution evenly in the simplified model.




WEDO · STEP 2; The outgoing links split A's contribution evenly in the simplified model. then C receives 0.10; final PageRank needs damping, all incoming contributions and repeated iterations.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain

C receives 0.10; final PageRank needs damping, all incoming contributions and repeated iterations.

IMPROVE: Explain one correction to your own reasoning.